

Healthcare Analytics and Hospital Operations

Intelligence Using Python and Power BI

1. Project Overview and Objective

This project focuses on analyzing healthcare and hospital operational data to identify important patterns, performance gaps, and areas for improvement. The project uses Python for data cleaning, pre processing, validation, and exploratory analysis, and Power BI for data modelling , visualization, KPI analysis, and interactive dashboard development.

The main objective is to demonstrate data pre-processing techniques using python and an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources

- Source: [Hugging Face Datasets](#)
- Year / Timeline: **Data collected during 2024–2025**

3. Problem Statement

- To analyse patient demographics and admission patterns across departments and wards.
- To evaluate health-condition and severity patterns among admitted patients.
- To assess hospital operational efficiency by analysing patient volume, staff availability, room availability across departments and wards.
- To analyse patient length of stay and estimated discharge dates to understand patient flow, and support hospital resource planning.
- To analyse insurance types and admission deposit patterns to understand financial aspects of patient admissions.
- To perform data cleaning, pre processing, validation and exploratory data analysis using Python.
- To develop an interactive Power BI dashboard presenting healthcare and hospital operational KPIs, trends, and actionable insights to support management decision-making.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Patient ID	Integer / String	Unique identifier for each patient
Admission Date	Date / Date time	Date on which the patient was admitted to the hospital
Age	Numeric (Integer)	Age of the patient in years
Gender	Categorical (String)	Gender of the patient
Department	Categorical (String)	Medical department handling the patient
Ward	Categorical (String)	Ward or unit where the patient is assigned
Admission Type	Categorical (String)	Mode of admission (e.g., Emergency, Urgent, Elective)
Severity	Categorical / Ordinal	Severity level of the patient's medical condition
Health Condition	Categorical (String)	Primary diagnosis or health condition of the patient
Staff Available	Numeric (Integer)	Number of healthcare staff available during admission
Available Rooms	Numeric (Integer)	Number of rooms available in the ward/facility
Visitors	Numeric (Integer)	Number of visitors allowed or recorded for the patient
Insurance Type	Categorical (String)	Type of health insurance held by the patient
Admission Deposit	Numeric (Float)	Initial deposit amount paid at the time of admission
Length of_ Stay Days	Numeric (Integer)	Total duration of hospital stay in days
Admission Year	Numeric (Integer)	Year component of the admission date
Admission Month	Numeric (Integer / String)	Month component of the admission date
Admission Day	Numeric (Integer)	Day of the month component of the admission date
Estimate _Discharge_ Date	Date / Date time	Projected date of discharge for the patient

5. Tools & Technologies

Based on the analysis project shown in your Google Colab notebook setup (covering descriptive, diagnostic, predictive, and prescriptive analysis for hospital patient and operational data), here are the core tools, libraries, and technologies used:

- **Core Programming & Environment**
 - * Python: Initial EDA, Data Cleaning ,Pre processing , EDA and Visualizations
 - * Jupyter Notebook / Google Colab: Interactive development environment (IDE) used for writing code, executing cells, and displaying visual outputs.
- **Data Processing & Numerical Computation**
 - * Pandas: Used for data manipulation, structured data cleaning, handling missing values, filtering hospital departments, removing duplicate values and calculating summary metrics (e.g., mean length of stay, total admissions).
 - * NumPy: Used for efficient numerical operations.

Exploratory Data Analysis & Visualization

- * Matplotlib: Core plotting library used to create charts
- * Seaborn: High-level statistical visualization library used for correlation heatmaps (e.g., severity vs. length of stay), box plots, and distribution charts across patient demographics.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

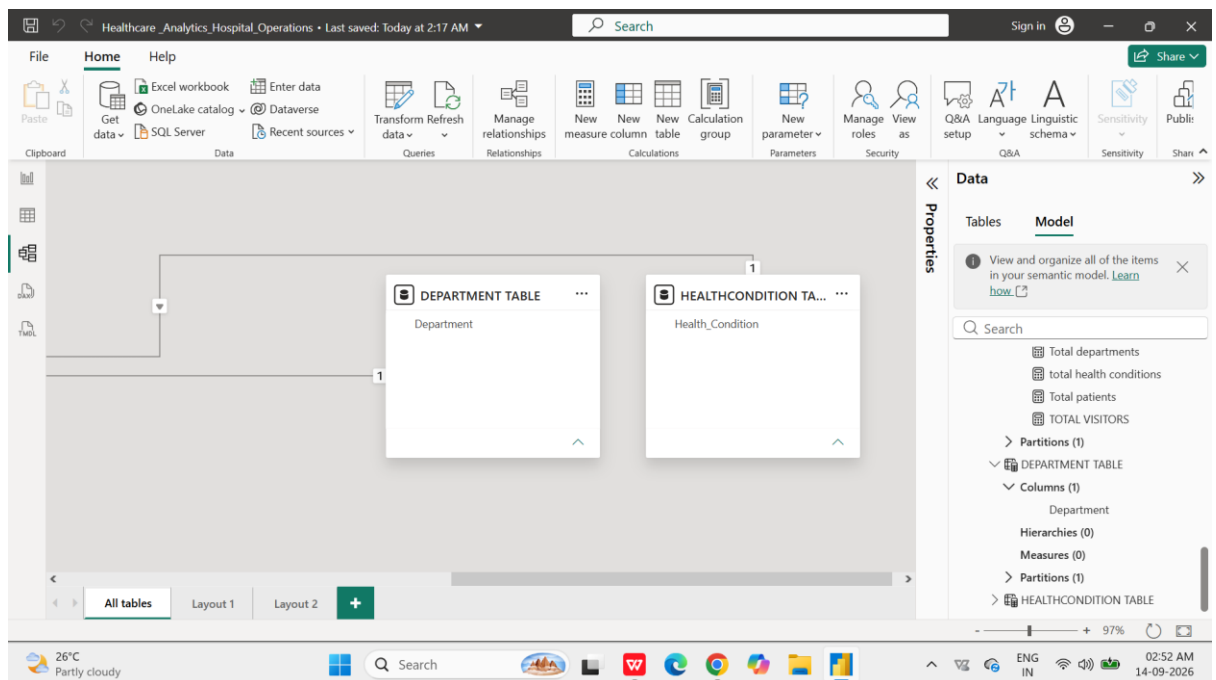
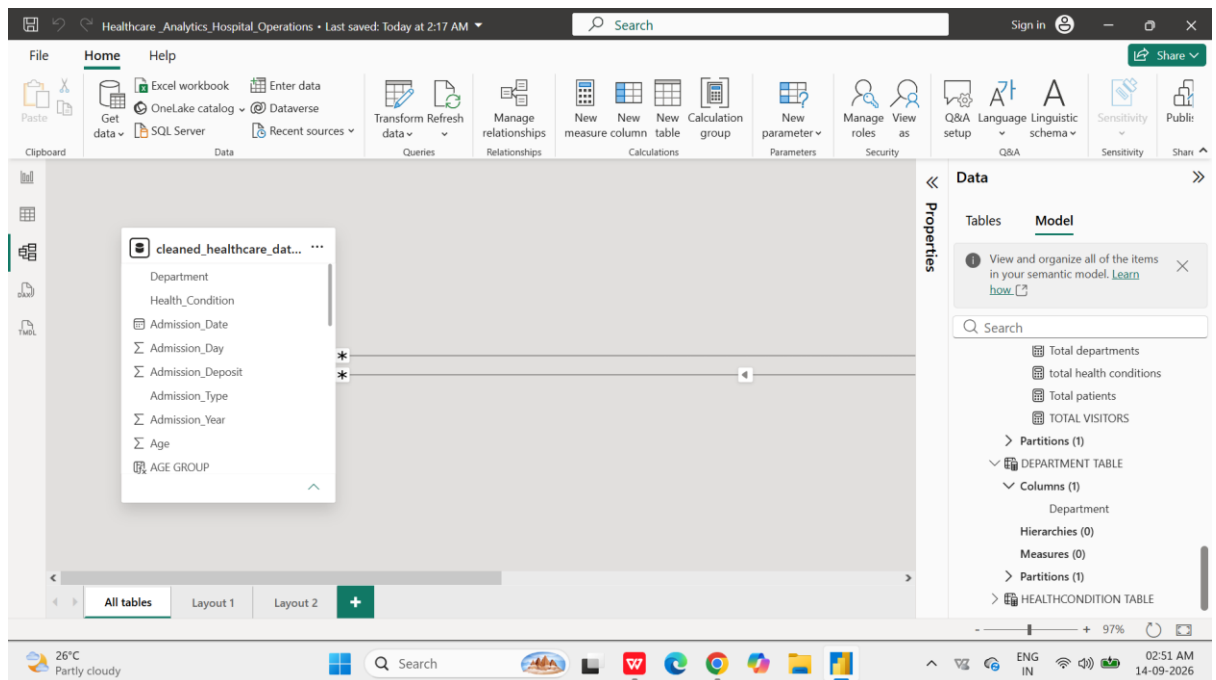
6. Data Pre-Processing (python / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:**
 - Removed duplicates,
 - handled missing values,
 - standardized formats,
 - created calculated fields.
 - Date type conversion
 - Text cleaning and standardization
 - Swekness check
 - Outlier detection by IQR model
 - Outlier handling
 - Final data validation
- **Filtering & Sorting:** Organized data to focus on relevant records.

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total admissions .
- Report interactive Layout using **Drill-down, Filters and slicers**

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File Home Help Table tools Measure tools

Name: Total patients Format: Whole number Data category: Uncategorized

Home table: cleaned_healthcare_data...

Structure: 1 Total patients = DISTINCTCOUNT('cleaned_healthcare_data 1'[Patient_ID])

Patient_ID	Admission_Date	Age	Gender	Department	Ward	Admission_Type	Severity	Health_Condition	Staff_Available	Available
P100118	12 April 2024	66	Male	Gynecology	Emergency	Elective	Mild	Diabetes	5	
P100174	12 June 2025	63	Male	Oncology	Emergency	Elective	Mild	Heart Disease	3	
P100534	27 July 2024	43	Male	Emergency	Emergency	Elective	Mild	Hypertension	25	
P100768	16 December 2024	73	Male	Oncology	Emergency	Elective	Mild	Arthritis	12	
P100849	22 August 2025	87	Male	Cardiology	Emergency	Elective	Mild	Hypertension	27	
P101257	11 June 2024	29	Male	Pediatrics	Emergency	Elective	Mild	Infection	7	
P101271	26 September 2024	40	Male	Emergency	Emergency	Elective	Mild	Heart Disease	14	
P101349	21 June 2025	40	Male	Oncology	Emergency	Elective	Mild	Diabetes	25	
P101410	26 July 2024	17	Male	Gynecology	Emergency	Elective	Mild	Infection	10	
P101807	07 January 2024	33	Male	Emergency	Emergency	Elective	Mild	Unknown	12	
P101974	06 September 2024	11	Male	Emergency	Emergency	Elective	Mild	Unknown	25	
P102011	29 December 2024	30	Male	Pediatrics	Emergency	Elective	Mild	Unknown	20	
P102127	12 August 2024	50	Male	Orthopedics	Emergency	Elective	Mild	Asthma	21	
P102170	16 December 2025	28	Male	Emergency	Emergency	Elective	Mild	Infection	25	
P102205	28 November 2024	80	Male	Gynecology	Emergency	Elective	Mild	Infection	18	
P102295	06 June 2025	65	Male	Pediatrics	Emergency	Elective	Mild	Heart Disease	18	
P102336	28 January 2025	79	Male	Neurology	Emergency	Elective	Mild	Unknown	30	
P102536	16 December 2025	61	Male	Neurology	Emergency	Elective	Mild	Infection	28	
P102713	18 November 2025	75	Male	Oncology	Emergency	Elective	Mild	Asthma	30	
P102912	07 March 2024	2	Male	Pediatrics	Emergency	Elective	Mild	Hypertension	16	

Table: cleaned_healthcare_data 1 (20,000 rows) Column: Total patients (0 distinct values)

Healthcare_Analytics_Hospital_Operations • Last saved: Today at 9:21 PM

File Home Help Table tools Measure tools

Name: TOTAL ADMISSION ... Format: Currency Data category: Uncategorized

Home table: cleaned_healthcare_data...

Structure: 1 TOTAL ADMISSION DEPOSIT (cr) = DIVIDE(SUM('cleaned_healthcare_data 1'[Admission_Deposit]),10000000)

Patient_ID	Admission_Date	Age	Gender	Department	Ward	Admission_Type	Severity	Health_Condition	Staff_Available	Available
P100118	12 April 2024	66	Male	Gynecology	Emergency	Elective	Mild	Diabetes	5	
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P102912	07 March 2024	2	Male	Pediatrics	Emergency	Elective	Mild	Hypertension	16	

Table: cleaned_healthcare_data 1 (20,000 rows) Column: TOTAL ADMISSION DEPOSIT (cr) (0 distinct values)

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File Home Help Table tools Measure tools

Name: Average staff Avail... Format: General Data category: Uncategorized

Home table: cleaned_healthcare... Auto

Structure Formatting Properties Calculations

1 Average staff Available = AVERAGE('cleaned_healthcare_data 1'[Staff_Available])

Patient_ID	Admission_Date	Age	Gender	Department	Ward	Admission_Type	Severity	Health_Condition	Staff_Available	Available
P100118	12 April 2024	66	Male	Gynecology	Emergency	Elective	Mild	Diabetes	5	
P100174	12 June 2025	63	Male	Oncology	Emergency	Elective	Mild	Heart Disease	3	
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P102912	07 March 2024	2	Male	Pediatrics	Emergency	Elective	Mild	Hypertension	16	

Table: cleaned_healthcare_data 1 (20,000 rows) Column: Average staff Available (0 distinct values)

27°C Light rain

Search

ENG IN 09:23 PM 12-09-2026

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File Home Help Table tools Measure tools

Name: AVERAGE LOS Format: General Data category: Uncategorized

Home table: cleaned_healthcare... Auto

Structure Formatting Properties Calculations

1 AVERAGE LOS = AVERAGE('cleaned_healthcare_data 1'[Length_of_Stay_Days])

Patient_ID	Admission_Date	Age	Gender	Department	Ward	Admission_Type	Severity	Health_Condition	Staff_Available	Available
P100118	12 April 2024	66	Male	Gynecology	Emergency	Elective	Mild	Diabetes	5	
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P102912	07 March 2024	2	Male	Pediatrics	Emergency	Elective	Mild	Hypertension	16	

Table: cleaned_healthcare_data 1 (20,000 rows) Column: AVERAGE LOS (0 distinct values)

27°C Light rain

Search

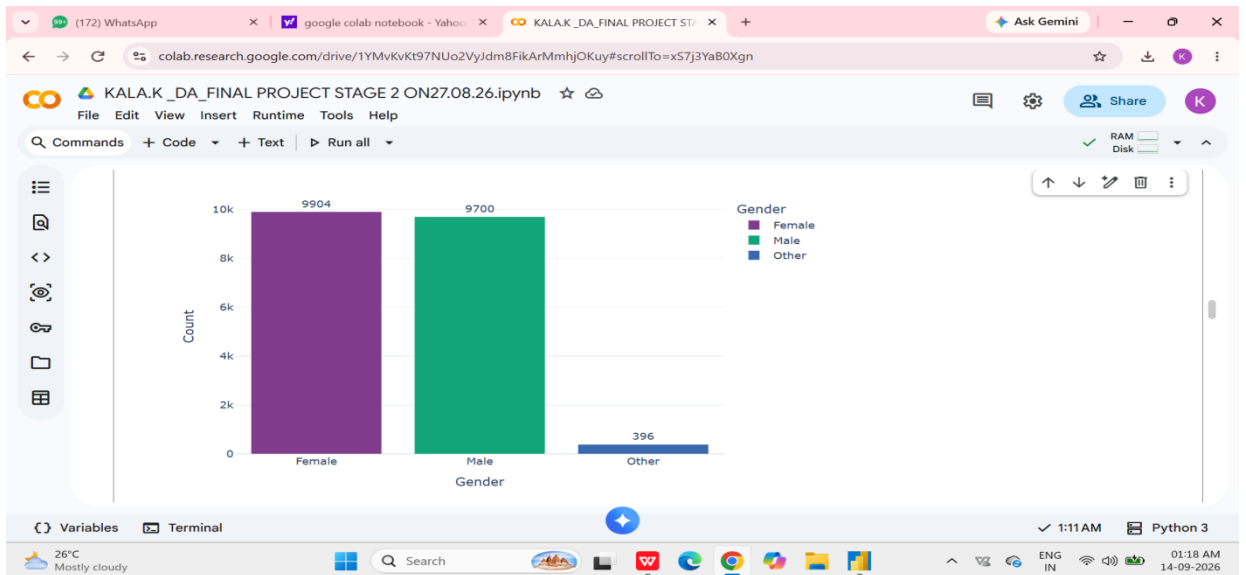
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8. Analysis and Visualizations (Power BI AND PYTHON)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** Univariate analysis charts, Bivariate analysis charts, multivariate analysis charts, KPI cards, and tables to communicate insights. **List are**
Count plot, histogram, box plot, scatter plot, bar plot, correlation heat map, pair plot, grouped analysis, pivot tables, advanced plot.

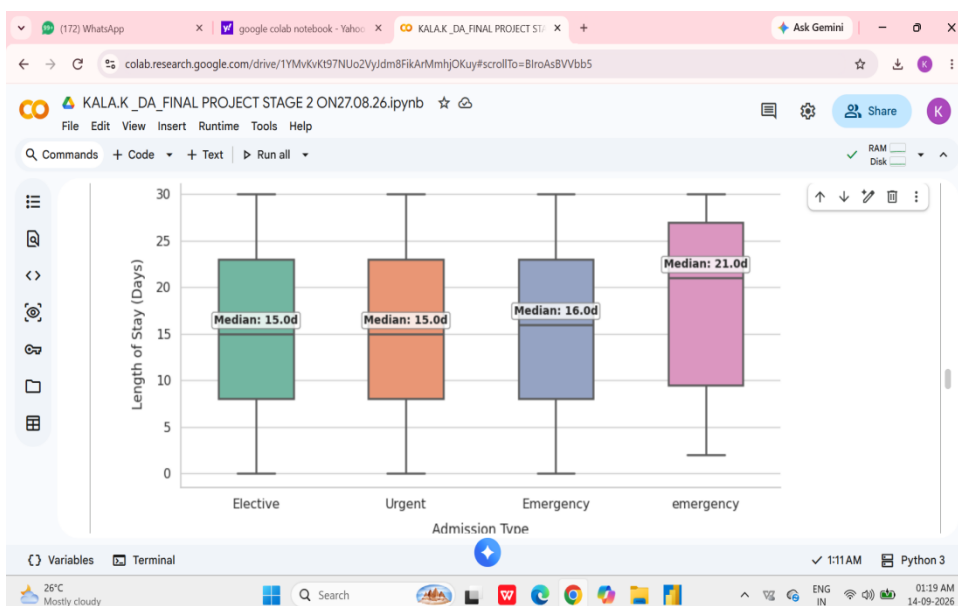
Patient distribution by gender



INSIGHTS Key Data Insights

- *Balanced Primary Demographics:* Patient admissions are almost evenly split between Female (9,904 patients) and Male (9,700 patients), showing balanced utilization across major demographic segments.
- *Inclusive Minority Representation:* The Other category accounts for 396 patients (~2% of total admissions), representing non-binary, gender-diverse, or non-disclosed patient profiles.

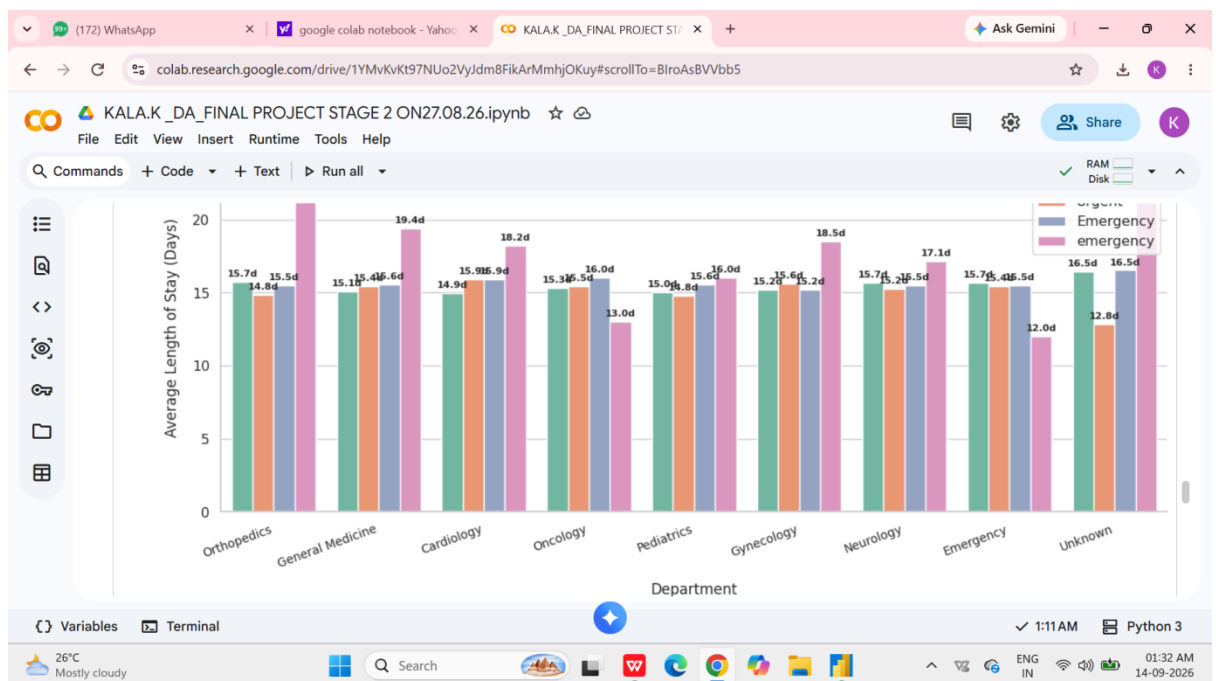
BIVARIATE ANALYSIS-LENGTH OF STAY BY ADMISSION TYPE



This bivariate box plot illustrates the relationship between admission urgency categories (Elective, Urgent, Emergency) and the hospital length of stay measured in days across 20,000 admission records.

- Admission type alone does not cause significant variance in stay duration, directing further analysis toward medical severity or department types.

MULTIVARIATE ANALYSIS :AVERAGE LENGTH OF STAY BY DEPARTMENT AND ADMISSION TYPE



This grouped bar plot compares the mean length of stay (in days) across various medical departments (such as Orthopedics, General Medicine, Cardiology, Oncology, Pediatrics, Gynecology, Neurology, Emergency, and Unknown), segmented by admission channel (Elective, Urgent, and Emergency)

The average length of stay remains constant at approximately 15 to 16 days across almost all combinations of clinical departments and admission channels.

The Unknown department category shows slightly more variation (with Urgent stays averaging ~13 days), but overall stay durations across major departments remain consistent.

- Report interactive Layout using **Drill-down, Filters and slicers**

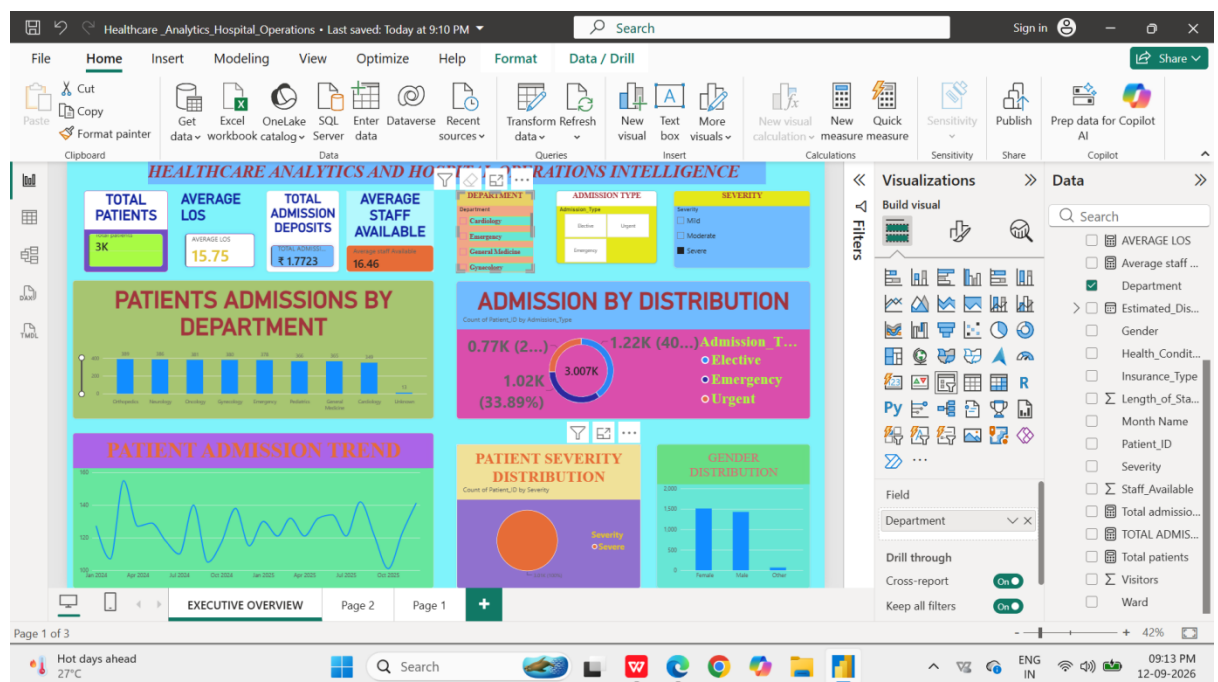
Interactive Dashboard Layout

The Power BI dashboard was designed with an interactive layout to enable users to explore healthcare operational data efficiently. Interactive features such as drill-down, slicers, and filters were incorporated to provide both high-level and detailed views of the data.

- **Drill-down,**
- **Filters**
- **Slicers**
- **Interative visuals , KPI, CARDS**

The interactive layout enables hospital administrators to move from overall performance to detailed analysis, compare different hospitals or departments, identify workload and performance variations, and obtain actionable insights for manpower planning, resource allocation, and operational improvement.

POWER BI DASH BOARD SCREEN SHOT



EXECUTIVE OVERVIEW PAGE -1

KPIS USED

- ✓ Total Patients- measures overall admission /admission volume -
- ✓ Average LOS-measures average duration of hospital stay
- ✓ Total Admission Deposits-shows total admission related deposit value
- ✓ Average Staff Available-indicates average staff availability associated with patient records

SLICERS USED

- ✓ Department slicer- can see patient volume, LOS, staff availability, admission pattern, severity, Gender distribution
- ✓ Admission Type slicer-elective, emergency, urgent
- ✓ SEVERITY SLICER -mild, moderate, severe

CHARTS USED

- ✓ Patients Admissions by Department

Chart -column chart

Usage

Shows the patient workload across departments.

Hospital Administrator Insight

The administrator can identify:

High-volume departments

Low-volume departments

Departments requiring greater staffing

Departments requiring more beds/resources

Workload imbalance between departments

Management decision

If one department consistently has substantially higher patient volume, management can consider: example emergency dept high volume, in elective at mild severity level, staff available average -16%

staff allocation → bed allocation → equipment allocation → duty roster planning.

- ✓ Admission Distribution

Chart - Donut chart

Usage

Shows the proportion of patients according to admission type.

Administrator Insight

This helps determine whether the hospital workload is predominantly:

Emergency-high

Urgent

Elective

Management decision

A high proportion of emergency admissions could indicate the need for:

Adequate emergency staffing

Faster triage

Emergency bed availability

Resource readiness

Appropriate shift planning

✓Patient Admission Trend

Chart -Line chart

Usage

Shows how patient admissions change over time.

Administrator Insight

The administrator can identify:

Increasing admission periods

Decreasing admission periods

Admission peaks

Low-demand periods

Possible workload fluctuations

Management decision

Admission trends can support:

staff roster planning + bed planning + inventory planning + operational preparedness.

✓Patient Severity Distribution

Chart -pie chart

Usage

Shows the distribution of patients according to clinical severity category.

Administrator Insight

This helps management understand the clinical workload, not just the number of patients.

For example: emergency

High patient count + high severity → greater clinical resource requirement.

High patient count + predominantly mild severity → different resource requirement.

Management decision

Severity distribution can support:

Staff deployment

Bed planning

Monitoring requirements

Nursing workload planning

Critical-care resource planning

✓Gender Distribution

Chart - column chart

Usage

Shows the patient distribution by gender.- female -high admission

Administrator Insight

It provides a basic demographic profile of the hospital's patient population.

This can support:

Service planning

Patient demographic analysis

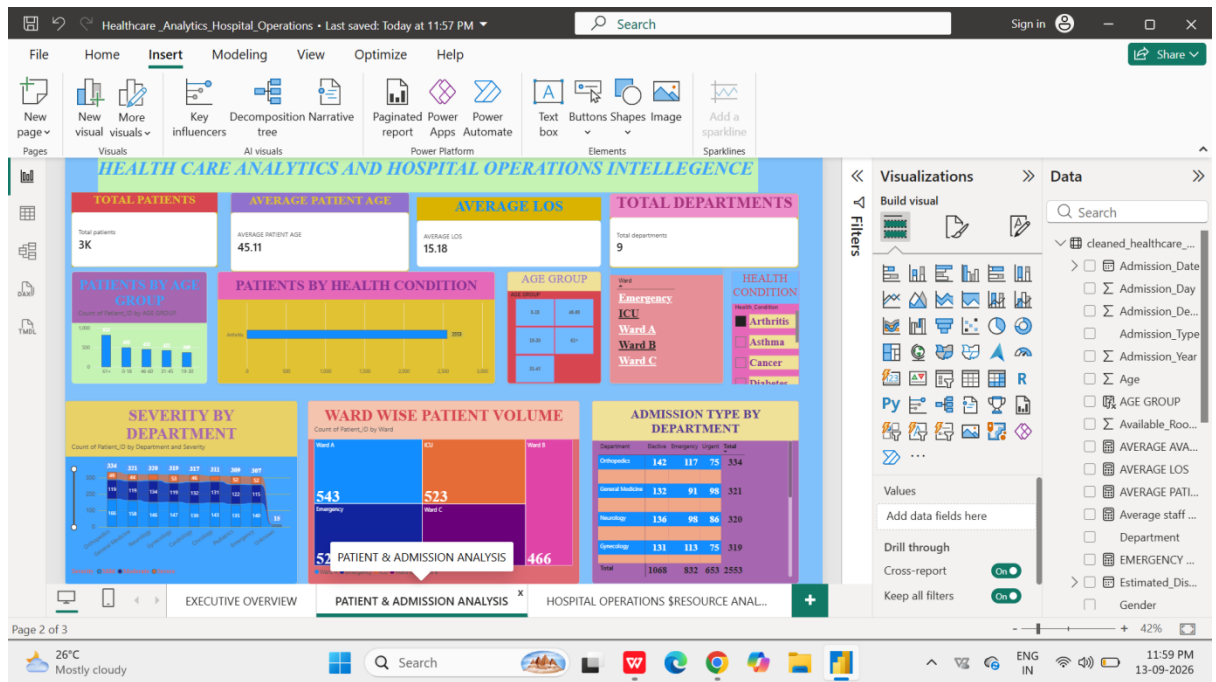
Department-level demographic comparison

Resource planning where gender-specific services are relevant

summary

"Page 1- an EXECUTIVE OVERVIEW . It provides a high-level view of patient volume, average length of stay, admission deposits and staff availability. Department and admission-type and severity slicers allow administrators to drill into specific operational segments. Department-wise admissions identify workload concentration, while the admission trend highlights changes in patient demand over time. Severity distribution provides insight into clinical workload, and gender distribution gives a basic demographic profile. Together, these visuals support data-driven decisions related to staffing, bed utilization, resource allocation and operational planning."

DASH BOARD 2 - PATIENT AND ADMISSION ANALYSIS



1. KPI CARDS

Total patients - over all patient volume

Average age -understand patient age profile

Average LOS-understand duration of hospitalization

Total Departments -shows breadth of department activity

2. SLICERS

Age group

0–15

16–30

31–45

46–60

61+

Usage

Allows the administrator to analyse patient characteristics by age group.

Ward

Emergency

ICU

Ward A

Ward B

Ward C

Usage

Allows the administrator to focus the entire page on a particular ward.

Health condition

Arthritis

Asthma

Cancer

Diabetes

Heart disease

Hypertension

Infection

Usage

Allows administrators to analyse a particular clinical condition.

CHARTS

1. PATIENTS BY AGE GROUP

Chart -column -understand age distribution

2. PATIENTS BY HEALTH CONDITION

Chart - Horizontal bar -identify major health conditions

3. SEVERITY BY DEPARTMENT

Chart -stacked column -identify clinical workload by department

4. WARD WISE EPATIENT VOLUME

Chart -Tree map- compare ward work load

5. ADMISSION TYPE BY DEPARTMENT

Chart-matrix-understands admission patterns by department

PAGE 2 - INSIGHTS

This is the most important part of your analysis report.

Insight 1 — Patient demographics

Patients by Age Group identifies the dominant age groups.

☐ Helps in patient-care and service planning.

☐ Insight 2 — Disease burden

Patients by Health Condition identifies which conditions contribute most to hospital workload.

☐ Helps prioritize clinical resources.

Insight 3 — Clinical workload

Severity by Department combines patient volume with severity.

☐ Helps identify departments requiring greater clinical attention.

Insight 4 — Ward workload

Ward-wise Patient Volume shows where patients are concentrated.

□ Helps with bed and staff allocation.

Insight 5 — Patient flow

Admission Type by Department identifies admission patterns across departments.

□ Helps with emergency readiness and operational planning.

Insight 6 — Focused analysis

The Age Group, Ward and Health Condition slicers allow administrators to investigate specific patient segments.

For example:

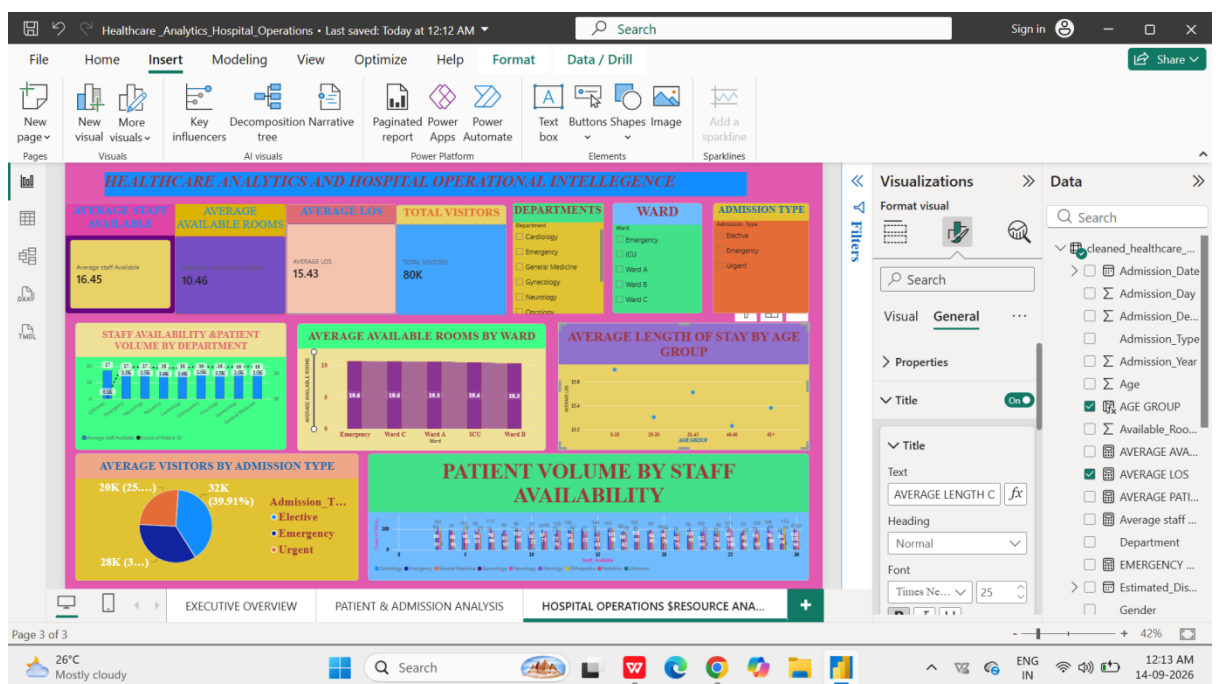
Select EMERGENCY + 61+ + Diabetes= 462

and the dashboard can show the corresponding patient distribution and admission characteristics, subject to the available data.

summary

“Page 2 PATIENT AND ADMISSION ANALYSIS focuses on patient and admission analytic. It analyses the patient population by age group and health condition and evaluates the distribution of clinical severity across departments. Ward-wise patient volume identifies differences in operational workload, while admission type by department provides insight into patient-flow patterns. Interactive filters for age group, ward and health condition enable administrators to perform focused analysis. These insights can support staffing, bed allocation, resource planning, emergency preparedness and workload management.”

DASHBOARD 3 HOSPITAL OPERATIONAL AND RESOURCE ANALYSIS



KPI Cards:

Average Staff Available

Average Available Rooms

Average Length of Stay

Total Visitors

Slicers:

Department

Ward

Admission Type

CHARTS

1. Staff Availability & Patient Volume by Department

Line and Clustered Column Chart

Compares staff availability and patient volume across departments to identify departments requiring further workload and resource review.

2. Average Available Rooms by Ward

Ribbon Chart

Compares average room availability across wards and supports ward-level capacity and resource planning.

3. Average Length of Stay by Age Group

Scatter Chart

Examines the relationship between patient age groups and average length of stay to identify patterns in hospitalization duration.

4. Average Visitors by Admission Type

Pie Chart

Shows visitor patterns across admission types and supports visitor-management and patient-support planning.

5. Patient Volume by Staff Availability

Line and Clustered Column Chart

Examines patient volume at different staff-availability levels and supports workload assessment and staffing review.

- **Create a consolidated Report /Dashboard.**

9. Insights

Key Insights

- Patient workload varies across staff and departments, indicating differences in workload distribution and staffing requirements.
- Staff availability has a direct impact on patient workload. Departments with higher patient volume but comparatively lower staff availability may require better staff allocation.

- Department-wise workload analysis helps hospital administrators identify high-demand departments and plan manpower accordingly.
- Hospital performance differs across facilities and measures, highlighting areas where corrective action and closer monitoring may be required.
- Interactive slicers allow administrators to analyse workload and performance by hospital, department, and other relevant categories.

10.Types of Analysis Used

Four major types of data analysis were used in this project to understand hospital operations and support data-driven decision-making.

1. Descriptive Analysis

Descriptive analysis was used to understand what has happened in the hospital data. It summarizes important information such as patient volume, staff availability, departmental workload, hospital performance, and key performance indicators (KPIs).

Examples: KPI cards, totals, averages, department-wise patient workload, and hospital-wise performance.

2. Diagnostic Analysis

Diagnostic analysis was used to understand why certain patterns or variations occurred in the data. Different departments, hospitals, staff availability, and workload levels were compared to identify possible operational gaps and areas requiring attention.

Examples: Comparing patient workload with staff availability, analysing departmental differences, and identifying performance variations.

3. Predictive Analysis

Predictive analysis was used to identify possible future trends and workload requirements based on the available historical data. The analysis can help

hospital administrators anticipate periods of increased workload and plan manpower and resources accordingly.

Examples: Observing workload trends and using historical patterns to support future staffing and resource planning.

4. Prescriptive Analysis

Prescriptive analysis focuses on what actions can be taken based on the findings. The insights generated from the dashboard were used to recommend appropriate actions such as improving staff allocation, monitoring high-workload departments, strengthening data quality, and optimizing resource utilization.

Examples: Recommendations for manpower allocation, workload management, resource planning, and continuous performance monitoring.

Overall Analytical Approach

The four levels of analysis—Descriptive, Diagnostic, Predictive, and Prescriptive—provide a structured approach from understanding the current situation to identifying causes, anticipating future requirements, and recommending appropriate actions. This approach helps convert healthcare data into actionable insights for hospital administrators and supports evidence-based decision-making.

11.SUMMARY

The project analysed healthcare operational data using Python and Power BI. Data cleaning, preprocessing, analysis and visualization were performed to examine patient workload, staff availability, departmental performance and hospital-level indicators. Interactive dashboards containing KPIs, charts and slicers were developed to provide a comprehensive view of hospital operations.

10. Conclusions

The project demonstrates that healthcare analytics can transform raw hospital data into actionable insights. The developed dashboard can support hospital administrators in staff allocation, workload management, resource planning, performance monitoring and evidence-based decision-

making. Overall, the project highlights the value of combining Python and Power BI to improve operational efficiency and support continuous improvement in healthcare services.

11. Recommendations

The integration of python and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting.

the following recommendations are proposed for improving hospital operations and supporting data-driven decision-making:

1.Optimize Staff Allocation

Hospital management should review patient workload and staff availability regularly and allocate staff according to workload requirements across departments.

2.Monitor High-Workload Departments

Departments with consistently high patient volumes should be closely monitored to prevent staff overburdening and maintain quality of patient care.

3.Improve Resource Planning

Hospital resources such as manpower, equipment, beds, and other operational facilities should be planned based on patient demand and workload trends.

4.Strengthen Performance Monitoring

KPI- based monitoring should be conducted regularly to identify departments or facilities requiring corrective and improvement measures.

5.Improve Data Quality

Missing, incomplete, or inconsistent data should be minimized through standardized data-entry procedures and regular data validation.

6.Monitor Workload Trends

Historical and current workload trends should be analyzed to anticipate periods of increased patient demand and plan resources proactively.

7.Use Dashboard-Based Decision Making

Hospital administrators should use the interactive Power BI dashboard for regular monitoring, comparison, and evidence-based operational decisions.

8.Conduct Periodic Review

The dashboard and performance indicators should be reviewed periodically so that emerging operational issues can be identified early and appropriate action can be taken.

9.Support Continuous Quality Improvement

The insights generated from the analysis can be used to identify gaps, implement corrective actions, and evaluate whether improvements have resulted in better operational performance.